

Unresolved Regions as Local Sources: A Local Incoming-Gate Framework for Nested-Space Inheritance

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Research draft — local incoming-gate certificate **OPEN**

Declared interval $I = S(1) + [0.12, 0.18]$. Source held fixed. State evolved.

Abstract

This research draft formulates a *local* incoming-gate framework in which an unresolved nested region acts as a source for two incoming gravitational constraints. A local observer retains a reduced block of the Dirac Hamiltonian, a retarded memory kernel, and complementary initial-state data. The incoming quantum kernel is obtained by transporting a fixed upstream source along a metric history,

$$C_\Sigma[g] = U_g C_{\text{up}} U_g^\dagger, \quad \delta C_{\text{src}} = 0.$$

The history class contains two axial functions that perturb the areal radius on one connected coordinate interval I of positive length. The symbols N and β label the incoming constraints associated with lapse and shift variations of the declared action; they are not fit parameters of the history. Closing both constraints on I , or excluding the whole class by a necessary relation, is the local gate. Laboratory records presently give the generated verdict OPEN: three of nine error-budget components are enclosed, six remain missing, and sampled nodal maxima lie above the existence tolerance 3×10^{-11} . Open is a documented intermediate status. The manuscript claims no global parent/child matching and no observational identification.

1 Unresolved regions as local sources

This article is a focused companion to the broader Nested-Space Cosmology manuscript, version 0.26.0 [1]. Its organizing inheritance postulate remains

$$\mathcal{T}_\Theta^* \mathbb{D}_\Theta = \mathbb{D}_\Theta. \quad (1)$$

The same operator law is proposed to be inherited across nested regions. The earlier operator, recursion and energy-transfer calculations are reused within their recorded domains. Here the question is narrower: a source-fixed local consistency test. This OPEN companion preserves that foundation; it does not replace the broader manuscript or establish continuing regeneration.

A local observer in one nested region does not resolve every degree of freedom that still influences the local field. Write a finite quadratic Hamiltonian in parent/child blocks,

$$H = \begin{pmatrix} H_{AA} & B \\ B^\dagger & H_{BB} \end{pmatrix}. \quad (2)$$

Eliminating the complementary block is ordinary linear algebra. The retained resolvent is the inverse of the Schur complement

$$S = K_A - B K_B^{-1} B^\dagger, \quad G_{AA} = S^{-1} = ((\lambda I - H)^{-1})_{AA}, \quad (3)$$

where $K_A = \lambda I - H_{AA}$, $K_B = \lambda I - H_{BB}$, and λ is a complex spectral parameter outside the relevant spectra. The Feshbach self-energy is

$$\Sigma_A(\lambda) = B(\lambda I - H_{BB})^{-1}B^\dagger. \quad (4)$$

The complementary region disappears from the local basis and leaves an effective operator on the retained block.

In time, the same elimination is not a memory kernel alone. The complementary initial state survives as an inhomogeneous source

$$i\dot{\psi}_A = H_{AA}\psi_A - i \int_0^t B e^{-iH_{BB}(t-s)} B^\dagger \psi_A(s) ds + B e^{-iH_{BB}t} \psi_B(0). \quad (5)$$

The integral is retarded. The last term carries $C_{BB}(0)$ and the initial cross block C_{AB} . Occupation data live in that remainder; they are not selected by $\Sigma_A(\lambda)$. Nested-space cosmology interprets this triple—reduced operator, retarded memory, retained initial state—as the local source associated with an unresolved nested region. Inside and outside are relational roles of the inheritance map $\mathcal{T}_\Theta$, not assignments of gauge charge.

The present article turns that interpretation into a declared local test on an incoming surface Σ . A metric history g transports a *fixed* upstream source to an incoming kernel $C_\Sigma[g]$. Two constraint residuals, generated by lapse and shift variations, are then asked to vanish on a connected interval I of positive length. The laboratory already owns the chart, the source inventory, the retarded evolution, and a nine-component error interface. What it does not own is a closed certificate. The generated local verdict occupying that interface is **OPEN**.

Statements below are labelled by kind: *assumption* (chart, source, history class, action inventory); *imported identity* (published mathematics in its original domain); *repository identity* (checked relation in this codebase); *interpretation* (the nested-space reading of those relations); *currently unproved closure* (the local incoming gate).

2 Notation

Symbol	Kind	Meaning
I	assumption	Connected incoming interval $I = S(1) + [0.12, 0.18]$
Σ	assumption	Incoming surface; coordinate z along Σ
$s = T - T_\Sigma$	assumption	Normal displacement from Σ
$w(z), U(z)$	assumption	Two axial history functions
$\chi(s)$	assumption	Smooth normal cutoff; equal to 1 near $s = 0$
N, β, a	assumption	Background lapse, shift, axial scale; held fixed
C_{src}	assumption	Upstream source covariance; $\delta C_{\text{src}} = 0$
$U_g, F[g]$	repository	Causal transport of source columns by g
$C_\Sigma[g]$	repository	Incoming kernel on Σ
$\mathcal{E}_N, \mathcal{E}_\beta$	repository	Incoming lapse and shift constraints
G_N, G_β	repository	Raw Gaussian source vertices
ϵ_B	certificate	Nine-component error bound, $B = N, \beta$
Θ	interpretation	Inherited dimensionless law

3 Declared local model

The calculation uses the already declared Kantowski–Sachs / Painlevé–Gullstrand incoming chart. The areal radius of the reference slice is one at the evaluation surface. Lapse N , shift β , and axial

scale a remain the background fields of that slice. The intrinsic incoming identification and the physical seam location are unchanged. Couplings $A, q, \Omega, \zeta, V_{\text{full}}$ remain locked. No additional action term is introduced. Einstein–Gibbons–Hawking–York, local seam contributions, and the Gaussian closed-time-path functional are each counted once.

The history class is the two-function radius family

$$\delta r = \chi(s)[s w(z) + s^3 U(z)/6], \quad s = T - T_\Sigma, \quad (6)$$

with a smooth normal cutoff $\chi(s)$ equal to one near $s = 0$. The functions w, U share an axial cutoff equal to one on I , with support inside $S(1) + [0.09, 0.21]$. These are coordinate domains for a local calculation. They are not a cosmological duration. The functions w and U are represented by Chebyshev coefficients times that cutoff. Base coefficient counts 8, 16, 32, 64 belong to the same class. Every normal and mixed derivative is the derivative of these two functions. Independent pointwise jets are not a surrogate for the surface functions.

For amplitudes in a sufficiently small neighbourhood of zero, and with w, U supported in a compact axial interval together with a compact radial window K about $\rho = 1$ strictly inside the trapped chart, the radius stays positive on compact K . Any initial slice strictly before the perturbation sees the reference geometry and the original upstream columns. Incident data on the upstream characteristic are those same columns, with vanishing preparation tangents.

The symbols N and β name the two incoming constraints obtained from the declared action by varying the lapse and the shift. They are not free amplitudes in (6), and they are not adjustable source parameters. The history varies areal radius; the constraints ask whether that variation, together with the evolved state, can cancel both lapse and shift residuals on I .

4 Source-fixed incoming state

Assumption. The same quantum state means the same upstream source preparation, not a covariance pinned to Σ independently of the metric.

Repository identity. For an allowed history the incoming kernel is the evolved restriction of the original source columns,

$$C_\Sigma[g] = U_g C_{\text{up}} U_g^\dagger = F[g] C_{\text{src}} F[g]^\dagger, \quad \delta C_{\text{src}} = 0. \quad (7)$$

Source selection stays fixed while the state on Σ evolves. All source channels, coherences, and cross-position kernels are retained. Source energies remain labels; they are not substituted as outgoing momenta $k = -E$ on evolved columns.

The causal chain is then

$$C_{\text{src}} \xrightarrow{U_g, \delta C_{\text{src}}=0} C_\Sigma[g] \longrightarrow (\mathcal{E}_N[g], \mathcal{E}_\beta[g])|_I. \quad (8)$$

A compact-support radius perturbation that is silent in the past produces a retarded incoming variation

$$\delta C_\Sigma(z, z') = \delta F_\Sigma(z) C_{\text{src}} F_\Sigma(z')^\dagger + F_\Sigma(z) C_{\text{src}} \delta F_\Sigma(z')^\dagger. \quad (9)$$

The restriction map at Σ has vanishing parameter derivative for this radius family, because the intrinsic metric and normal on Σ are held fixed. The kernel on Σ nevertheless changes, because the Dirac evolution from the upstream source feels the history.

A frozen incoming C_0 is a different law. In the smooth compact class with exact prepared $C_\Sigma = C_0$, the incoming order-four geometric germ is forced to the reference germ, and the included shift constraint cannot close: the baseline Killing current satisfies the strict bound $P_K < -M$ with

$$M = \frac{5887634380411363}{265464355000000000} > 0. \quad (10)$$

That non-existence result remains a regression of the frozen class. It is not the active local gate under (7).

5 Lapse and shift constraints

Currently unproved closure. Both incoming constraints are required to close on one connected interval I of positive length, or a necessary relation must exclude the entire declared local history class, under the source-fixed evolved state (7).

The owned assembly is

$$\mathcal{E}_B[g] = \mathcal{E}_B[g_{\text{ref}}] + \Delta \mathcal{E}_B^{\text{local+ref}}[j_\Sigma g] + (G_B[C_\Sigma[g]] - G_B[C_{\text{ref}}]), \quad B = N, \beta. \quad (11)$$

Baseline and geometry are added once. The computational reference of each family is the stationary phase of the same initial fields; its history tangent vanishes. Current matter uses the weighted exact-phase pair (F, F_z) and the retarded tangents $(\delta F, \delta F_z)$.

Let F denote evolved canonical source columns weighted once with the original square roots of $dE/(2\pi)$, and let C_{src} be the fixed full source matrix. With axial derivative F_z ,

$$D = FC_{\text{src}}F^\dagger, \quad J = \frac{F_z C_{\text{src}} F^\dagger - F C_{\text{src}} F_z^\dagger}{2i}. \quad (12)$$

The raw Gaussian vertices, with inherited signed-family multiplicity μ , are

$$\begin{aligned} G_N &= \mu \left[m \text{Tr}(\sigma_1 D) - \frac{\ell}{r} \text{Tr}(\sigma_2 D) - \frac{1}{a} \text{Tr}(\sigma_3 J) \right], \\ G_\beta &= \mu \text{Tr} J. \end{aligned} \quad (13)$$

Only in the stationary reference does $F_z = -iEF$. That substitution is wrong for evolved columns and their tangents.

The local/reference geometric response on the compatible slots $(w, w_z, w_{zz}, w_{zzz}, U, U_z)$ has the certified principal matrix

$$\begin{pmatrix} \partial_U \mathcal{E}_N & \partial_{w_{zz}} \mathcal{E}_N \\ \partial_{U_z} \mathcal{E}_\beta & \partial_{w_{zzz}} \mathcal{E}_\beta \end{pmatrix} = \begin{pmatrix} c_u & B \\ d & e \end{pmatrix}, \quad (14)$$

with $c_u > 0$ from the lapse variation of the third normal radius derivative and c_v the mixed-normal shift coefficient in $\mathcal{E}_\beta = S_\beta + c_v \delta r_{Tz}$. These coefficients are action responses. They are not new couplings. Geometric invertibility of (14) is a useful preconditioner. It is not existence in the coupled state class.

A closed local result is one of the following.

- **Local EXISTENCE:** a verified history g in the declared class with both residuals and the nine-component error budget inside the componentwise tolerance

$$\sup_{z \in I} (|\hat{\mathcal{E}}_B(z)| + \epsilon_B(z)) \leq 3 \times 10^{-11}, \quad B = N, \beta. \quad (15)$$

- Scoped local NON-EXISTENCE: a necessary relation covering the entire declared class, with a strict error-controlled separation from zero. Failed optimization is not non-existence.

The generated placeholder occupying the result slot is **OPEN**.

6 Imported mathematics

The local calculation reuses published mathematics inside the domains stated by those authors. Application to nested-space cosmology requires the connection that is actually shown.

Klich [2] proves a Fock-space trace identity that reduces certain traces of quadratic fermionic operators to single-particle determinants, and thereby recovers Levitov’s full-counting-statistics formula. The published chapter is a contribution to mesoscopic quantum noise. The identity is used here as established algebra for a Gaussian fermionic functional. It does not select an occupation, close (15), or supply a cosmological source.

Gérard, Häfner and Wrochna [3] construct the Unruh state for massless Dirac fields on slowly rotating Kerr spacetimes. In their stated Kerr–Kruskal domain, Theorem 1.1 proves that the state is asymptotically thermal at Hawking temperature on the past event horizon and is pure and Hadamard in the union of exterior and interior regions. Affine-frequency horizon data and incoming-vacuum data at past null infinity belong to that theorem. The theorem does not automatically prove the characteristic traces, spin/current maps, or Hadamard hypotheses on the nested-space black-universe chart used here.

Bronnikov, Dehnen and Melnikov [4] describe regular black-universe geometries: an exterior resembling Schwarzschild, a regular minimum of areal radius, an expanding Kantowski–Sachs interior, and a de Sitter infinity in place of a singularity. Those metrics and causal extensions are reusable geometric examples. Their supporting phantom source is not the selected Dirac / closed-time-path source of the local gate.

Blanes and Moan [5] construct commutator-free Magnus integrators. Equation (43) of that paper is the fourth-order two-exponential scheme

$$x_h = \exp(\alpha_1 h A_1 + \alpha_2 h A_2) \exp(\alpha_2 h A_1 + \alpha_1 h A_2) x_0 \quad (16)$$

used as a numerical method on the extended linear system when a time integrator of that family is compared. Formal order is not an error certificate for (15). Production evaluations use the owned DOP853 path; CF4 is a comparison method with its own implementation controls.

7 Repository identities and interpretation

The compact local-observer correspondence checks five finite identities independently of the incoming-gate search. The owner and record paths are listed in Section 9.

Imported algebra, repository-checked. The joined resolvent equals the ordered Schur complement (3). Reversed multiplication, dropping $B^\dagger$, replacing the child resolvent by $BB^\dagger/\lambda$, and transposing the coupling each disagree with the joined inverse.

Repository identity. Eliminating the complementary region in time retains both the memory integral and the initial-state term in (5). Omitting $\psi_B(0)$ changes the resolvent solution. The retarded self-energy is independent of occupation; the noise term is not. The same Schur map has vanishing vacuum current and a nonzero excited current.

Repository identity. The same Hamiltonian owns conservation and metric variation,

$$\dot{E}_A = \text{Tr}(C J_A) + \text{Tr}(C \dot{h}_A), \quad \dot{E}_{\text{tot}} = \text{Tr}(C \dot{H}), \quad \text{Tr}([H, C]H) = 0, \quad (17)$$

and, on the physical history at fixed preparation,

$$\left. \frac{\delta \Gamma_F}{\delta X^A} \right|_{X_+ = X_-} = - \text{Tr} \left(C \frac{\delta H}{\delta X^A} \right). \quad (18)$$

The child's Gaussian determinant is counted once. Omitting the complementary Schur variation fails the finite check.

Repository identity. Normalized inheritance on a finite chain,

$$\Gamma_n(x) = xI - H - \frac{1}{\Omega} b \Gamma_{n+1}(x/\Omega)^{-1} b^\dagger, \quad (19)$$

agrees with the direct multi-room inverse. Dropping $1/\Omega$ or reversing a noncommuting link fails. The control does not select a physical Ω .

Interpretation. A local observer in one nested region is described by the reduced block, the retarded memory, the retained initial-state and noise data, and the same-action variation of the remaining Hamiltonian. Sheet exchange in the tested two-sheet Dirac embedding is a distinct operator from charge conjugation; it is not antimatter.

Currently unproved. Those identities do not close (15). They supply the language in which the local gate is posed.

The homogeneous frequency-diagonal Kantowski–Sachs class without interface is already decided under a different truncation: the serialized state produces a nonzero shift residual, while every owned homogeneous local/reference term has vanishing axial momentum, so no smooth stationary history exists in that class. That certificate is not transferred to the two-function incoming family (6).

8 Numerical method, inventory, and present evidence

The production evaluator evolves each candidate from fixed upstream batches and returns both constraints together with all retarded history derivatives. Source parameters, coherences, weights, and the seam remain immutable. Cache identity binds the analytic profile, the computational mesh, interpolant degree, solver options, and channel labels. Amplitudes alone are not a cache key.

A damped Newton or Gauss–Newton search, when used, carries the full retarded Jacobian. The certified geometric principal operator (14) may be used as a preconditioner. A nodal hit is not a continuous residual bound on I .

The retained source inventory materializes authenticated panels with original quadrature weights, signed angular partners, and the three $\ell = 0$ groups that keep their baseline allocation because their radius-history response vanishes identically in this family. Contour bilinears are not accepted as physical Gaussian mode columns. Loading the inventory is not a source certificate.

Current records, read from the source-bound note and associated development JSON named in Section 9, are as follows. The physical column-accuracy universe is 164 panels / 49372 rows. An older 102-panel low/subgap subset inspects 27828 rows below the joined V5 thresholds, 1528 rows at or above, and 59 straddling panels. Those row counts classify coverage. They are not a directed N, β bound.

Certificate v2 requires nine nonnegative componentwise error bounds on I , added to a continuous residual upper bound, before an existence decision. The v4 budget records three enclosed or reused components and six missing components. The partial known error sum, rounded for display, is

$$(8.99915 \times 10^{-12}, 6.99608 \times 10^{-14}) \quad (20)$$

for (N, β) . That pair is a sum of presently owned pieces. It is not a closure bound. The successor 257-node budget keeps all nine slots open until applicable bounds are authenticated; the 129-node phase result cannot be transferred by relabelling it.

Component	N	β
Covered baseline regions	8.91625×10^{-12}	3.88237×10^{-18}
Phase value	8.28992×10^{-14}	6.99570×10^{-14}
Energy interpolation	4.83154×10^{-26}	3.78403×10^{-26}
Field / space / time (whole-cone)	missing	missing
Changed-history ultraviolet tail	missing	missing
Baseline low / subgap	missing	missing
Upstream continuation	missing	missing
Between-node remainder	missing	missing
Arithmetic remainder	missing	missing
Partial known sum	8.99915×10^{-12}	6.99608×10^{-14}

Covered-region reuse applies only to unchanged V5 baseline spectral regions. Phase-value enclosure covers 129 nodal phase values, including phase-contraction arithmetic, and excludes the between-node remainder. Energy interpolation is an ideal difference-interpolation remainder; node solves and arithmetic are excluded. Whole-cone field work restores directed residual integrals on the backward cone of I . A restored one-cell comparison on family 14₁, cell 122, stays inside its recorded directed enclosure. That comparison is one cell. It does not supply a full-cone bound, and it does not replace physical $\rho = 1$ source error.

The reanchor record reports candidate nodal maxima (rounded for display)

$$(9.64093 \times 10^{-4}, 3.80960 \times 10^{-4}) \quad (21)$$

for (N, β) on 129 nodes, for the current measured profile in the declared class. Those numbers are sampled maxima, not continuous suprema, and not existence residuals in the sense of (15). They lie orders of magnitude above 3×10^{-11} . They are not compensated by (20).

An earlier discrete coupling control of the evolved-constraint assembly reports residual $1.352287829051212 \times 10^{-9}$ against tolerance 3×10^{-8} . That check verifies that matter and the retarded derivative enter both constraints. It is not a physical root.

9 Reproduction

The public draft carries a selected snapshot of original laboratory records and two inspectable mathematical owners. Each byte is bound to a source Git blob and laboratory commit in `paper/local-gate-evidence/snapshot.json`. This snapshot supports inspection of the quoted checkpoint; it is not the complete dependency graph needed to replay the active scientific calculation. The historical public evidence chain remains independently reproducible.

The following commands, from the public repository root, authenticate the draft or rebuild it without starting scientific campaigns:

```
python scripts/verify_local_gate_draft.py
python scripts/build_local_gate_draft.py --check
```

The first checks source, record, PDF, and source-archive hashes. The second requires the compiler recorded in the build manifest, performs two clean builds, and requires identical PDF and bibliography bytes. Numerical values in the budget table and candidate pair are generated from the original records. A digest establishes artifact identity, not the truth of a physical claim. Closure still requires a fresh history evaluation and all nine applicable error bounds.

10 Limitations

Every claim in this article is restricted to the declared local incoming gate on I under the state law (7) and the history class (6). Global parent/child matching, extended stationarity, metric timestep, and observational comparison are later research. They are not hidden completion conditions.

The generated verdict is OPEN. Open is a documented intermediate status. Until all nine error components and a continuous residual upper bound are present, the existence criterion cannot fire. Until a class-wide necessary relation with strict error-controlled separation is present, the non-existence criterion cannot fire. No residual quoted from a control history is a physical root.

This article is a public research draft. It is not a closed-result release or an arXiv submission.

A Provenance

The source snapshot names laboratory commit `aefbdbaf` (full identity in the manifest) and the public base commit. The public Git history identifies subsequent editorial changes. The build manifest records source and PDF hashes, compiler identity, and the two-build comparison. The retained notebook v0.26.0 is unchanged.

The compiler used for this draft is identified exactly in the manifest. It is not an immutable arXiv build-container lock. Reproducible bytes on that runtime do not prove compatibility with arXiv’s current service. No arXiv license has been selected and no submission action is recorded.

Significant AI-use disclosure

OpenAI Codex and xAI Grok assisted with implementation, numerical workflow review, and language preparation. Douglas Ek is the sole author and assumes responsibility for every scientific claim. The local gate remains open.

References

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